The examination is being carried out on the **following application documents**

Description, Pages

1-11 as originally filed

Claims, Numbers

1-15 as originally filed

Drawings, Sheets

1/3-3/3 as originally filed

1 References

Reference is made to the following documents; the numbering will be adhered to in the rest of the procedure:

D1 US 2004/184710 A1 (KUBBY JOEL A [US] ET AL) 23 September 2004

D2 Peters Tjitte-Jelte ET AL: "Electrothermal Actuators for SiO2 Photonic MEMS",
Micromachines, vol. 7, no. 11, 7 November 2016 (2016-11-07), page 200, XP093144506,
ISSN: 2072-666X, DOI: 10.3390/mi7110200, URL:https://pure.tudelft.nl/ws/portalfiles/portal/8089350/micromachines_07_00200_v2.pdf

D3 Sharma Suraj ET AL: "Integrated 1x3 MEMS silicon nitride photonics switch",
Optics Express, vol. 30, no. 12, 6 June 2022 (2022-06-06), page 22200, XP093145083, ISSN: 1094-4087, DOI: 10.1364/OE.460533

D4 US 11 320 718 B1 (MAHMOUD MOHAMED [US] ET AL) 3 May 2022

D5 WO 2010/082524 A1 (NEC CORP [JP]; TAKAHASHI MORIO [JP]) 22 July 2010 (2010-07-22)

D6 US 2019/369328 A1 (DAVIES SAMUEL [GB] ET AL) 5 December 2019 (2019-12-05)

2 Independent claims 1 and 15 - Lack of novelty

2.1 The present application does not meet the requirements of Article 52(1) EPC because the subject matter of claims 1 and 15 is not new within the meaning of Article 54(1) and (2) EPC.

2.2 Document D1 discloses (the references in parentheses applying to this document, see figs. 1-5) all features of claim 1:

A structure (variable optical attenuator 10) comprising:

- a semiconductor substrate (silicon substrate 60 in fig. 3; see paragraph 30);
- a first dielectric layer (buried oxide 56 in fig. 3, see paragraph 30) on the semiconductor substrate, the first dielectric layer including a first cavity (the free space between the two left-most portions 56 in fig. 3) and an edge (from paragraph 30 and figs. 1-3, 5, it is clear that the oxide layer is removed to the left of the dashed line around optical slab 40, one edge being formed left to right near 58 in fig. 1) defining a boundary of the first cavity;
- an edge coupler including a waveguide core (the two waveguide cores 52, 54 on a cantilever that couple with stationary waveguides 42, 44), the waveguide core including a portion that extends past the edge of the first dielectric layer and overhangs the first cavity (the portions of the waveguide cores that are not fixed to the slab at 58); and
- a first heater (actuating beams 21 to be heated by current between ends 26 and 45, see paragraph 34) positioned adjacent to the portion of the waveguide core, the first heater spaced by a first gap (see fig.s) from the portion of the waveguide core.

2.3 The same reasoning applies, mutatis mutandis, to independent method claim 15 the subject-matter of which corresponds to that of to claim 1, and which is therefore also not considered to be new.

3 Dependent claims - Objections

The dependent claims do not appear to contain any additional features which, in combination with the features of any claim to which they refer, meet the requirements of the EPC with respect to novelty and/or inventive step, for the following reasons:

3.1 The additional features recited in the following claims are also anticipated:

- claims 2, 8, 9, 10: D1 (figs. 1, 3).
- claim 6: D1 (paragraph 34).

The subject matter of these claims is therefore also not new within the meaning of Article 54(1) and (2) EPC.

3.2 Regarding the additional features recited in

- claims 3, 4, 11: Adding a second heater is an obvious modification, see e.g. D2 (two sets of heatable beams in fig. 2); or D3 (two laterally spaced switching actuators in figs. 1, 2, 5); or D4 (symmetrical actuating electrodes 106 in fig. 1A).
- claim 5: Coupling the waveguide to a laser device is obvious see for example D5 (fig. 8).
- claim 7: Adding a passivation dielectric over the heater and the waveguide is an obvious alternative in light of D6 (passivation dielectric 609 in fig. 6).
- claim 12: Obvious design alternative, see for example D3 (fig. 5B).
- claim 13: A taper is obvious in light of D3 (fig. 2g).
- claim 14: D1, paragraph 30, discloses the waveguides and heater to be made from monocrystalline silicon, making the heater from tungsten is an obvious alternative, see for example D5 (paragraph 13).

The subject matter of these claims therefore cannot be considered to involve an inventive step according to Article 52(1) and Article 56 EPC.

4 Formal issues

4.1 The features of the claims should be provided with reference signs placed in parentheses to increase the intelligibility of the claims (Rule 43(7) EPC).

4.2 To meet the requirements of Rule 42(1)(b) EPC, the relevant documents cited above should be identified in the description and the background art disclosed therein should be briefly discussed. Care should be taken during revision, especially of the introductory portion and of any statements of problem or advantage, not to add subject-matter which extends beyond the content of the application as originally filed (Article 123(2) EPC).

4.3 If present, general and vague statements in the description ("spirit of the invention"/"all equivalents"; combinations of features; claim-like clauses) should be deleted (see Guidelines F-IV, 4.4).

4.4 If present, the formulation "incorporated by reference" used in the context of cited prior art documents should be deleted (see Guidelines F-III, 8).

5 Conclusion

5.1 Given the objections raised above, it is not at present apparent which part of the application could serve as a basis for a new, allowable claim. Should the applicant nevertheless regard some particular matter as patentable, amended claims should be filed.

Regarding an amended application the applicant should also consider the following points:

5.2 In order to facilitate the examination of the conformity of the amended application with the requirements of Article 123(2) EPC, the applicant should clearly identify the amendments made, irrespective of whether they concern amendments by addition, replacement or deletion, and indicate the **passages of the application as originally filed** on which these amendments are based (Rule 137(4) EPC).

5.3 Arguments as to why the independent claims are inventive should be provided by using the Problem-and-solution approach (Guidelines G-VII, 5). This requires explicitly indicating

a. the differences vis-à-vis the closest prior art,

b. the technical effect achieved by the differences,

c. the objective technical problem solved by the invention,

d. the reasons why the differences are neither obvious by themselves nor obvious in view of the teachings of the other cited prior art documents.